

YouTube Analytics Project

IE7945 / Master's Project | Northeastern University | Team SpaceMSDA

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Abstract—Executive Summary: The purpose of this project is to determine what drives the YouTube algorithm. Our team defines “YouTube success” as views, subscribers, and likes for your channel content. Specifically, this paper focuses on the short video promotion algorithm and not the long video algorithm. In addition, we will go in depth for the dataset window and sources, the main analytic methods (EDA, modeling, experiments), and the most important measurable outcomes (e.g., subscriber growth, CTR, retention, view velocity). We conclude that the YouTube algorithm imposes a hard limit on the number of views that the Short Feed promotes per short video. The limit is based on typical maximum views per short.

Index Terms—YouTube analytics, growth strategy, content effectiveness, machine learning, A/B testing, engagement

I. INTRODUCTION

The purpose of this project is to demonstrate our ability to apply analytics to a new YouTube channel for growing the number of views and subscribers. Since milestone 3, our team has done more A/B testing with interesting result to share. The result demonstrates the promotion of shorts by the algorithm and the limits imposed by the YouTube short algorithm. Since October 31 was Halloween, we targeted this holiday for A/B testing. Inside the report in the analysis section you will find comparison of A/B test set such as “Sun’s Halloween Face – Real NASA Image” and “Sun’s Solar Eruption – Real NASA Image”, which was both released on October 30 at 3 PM PT. Another test set with staggered release is “13 Milliseconds to a Black Hole (NASA Simulation)” and “NASA’s 5 Spookiest Nebulae – Your pick?”. The second set test the use of Emoji and numbers. The final 3 sets of A/B testing focused on Title changes for Question style versus Statement style.

At the time of this report, our YouTube Channel has gotten over 30k views, 441+ likes and 30+ subscribers. GitHub is updated with scraping scripts for data, modeling, and drives our content creation.

After initially analyzing the public Kaggle Trending Videos dataset, we focused training on a dedicated multimodal deep learning model for YouTube Shorts in the space-science niche. This model combines SentenceTransformer text embeddings, CLIP-based thumbnail embeddings, and engineered temporal features over 9,247 Shorts (Jan. 2022–Oct. 2025). Its primary purpose is not only to predict “winning” Shorts ($\geq 1,000$ views) but, more importantly, to quantify which content attributes matter most for Shorts performance—title semantics, thumbnail visuals, and timing signals.

Our main analysis uses a space- and science-focused dataset of 10,387 YouTube videos collected via the YouTube Data API (Jan. 2022–Oct. 2025). We build a comprehensive machine learning pipeline that predicts video log growth rate from 92 engineered features spanning temporal patterns, video characteristics, text signals, engagement metrics, and TF-IDF components. The pipeline evaluates seven algorithms (RandomForest, GradientBoosting, XGBoost, LightGBM, ExtraTrees, AdaBoost, and a simple neural network) and identifies LightGBM as the best-performing model with a test R^2 of 0.7320, providing a strong basis for interpreting which factors drive video performance in the space/science niche.

II. DATA ACQUISITION, CLEANING, AND PREPARATION

A. Data sources

We used Kaggle for our initial historical analysis (see EDA section), which we found limiting. For main

analysis and channel analysis, we used the YouTube API for scraping search results along with live channel data analysis. In addition, we can see Google Trends for keyword selection. The two combined will allow us to connect the dots for the new channel and drive content creation. Throughout our report we have labeled Kaggle (large, popular, general) vs. YouTube API (Space + science) as the data source as these two sets were analyzed separately.

We collected 10,387 YouTube videos using the YouTube API, filtered by space- and science-related keywords between January 2022 and October 2025 for main analysis. For each video we obtain publication timestamps, view, like, and comment counts, duration, titles, descriptions, and tags.

B. Data cleaning

The cleaning pipeline enforces logical constraints such as $\text{views} \geq \text{likes} \geq \text{comments}$ and removes records that violate these relationships. For example, 52 videos with likes exceeding views are dropped as logically inconsistent. We also remove extreme outliers in view counts using the 99.5th percentile (videos with more than 16.7M views) to prevent a small number of viral videos from dominating model training. Missing descriptions and tags are replaced with empty strings, preserving the information that a video lacks these fields instead of imputing synthetic content.

C. Feature engineering

We engineer 92 features across five categories: temporal (10 features), video-specific (4), text-based (20+ structural and pattern-detection metrics), engagement (3), and TF-IDF components reduced from 100 to 20 dimensions via SVD. Temporal features capture publication hour, day-of-week, month, quarter, days since published, and weekend indicators. Video-specific features include duration buckets (Short, Medium, Long) and log-transformed duration. Text features cover title and description length, word counts, capitalization ratios, and pattern detectors for questions, exclamations, numbers, emojis, and special characters, as well as counts of URLs, hashtags, and calls to action.

The target variable is the log-transformed growth rate,

$$\log_growth_rate = \log \left(\frac{\text{view_count}}{\text{days_since_published}} + 1 \right),$$

which normalizes for age, reduces skewness (from 18.7 to 0.3), and aligns with YouTube’s focus on view growth rather than absolute size.

D. YouTube Shorts dataset for winner prediction

To complement the Trending Videos analysis, we constructed a separate Shorts dataset focused on space-science content. The dataset contains 9,247 YouTube

TABLE I
CLASS DISTRIBUTION FOR THE SHORTS WINNER-PREDICTION TASK.

Dataset	Non-winners	Winners
Training	3,027 (42.0%)	4,181 (58.0%)
Test	763 (42.3%)	1,040 (57.7%)

Shorts uploaded between January 2022 and October 2025. Overall, the Shorts winner-prediction dataset is reasonably balanced. In the training split, 3,027 Shorts (42.0%) are non-winners with fewer than 1,000 views, while 4,181 Shorts (58.0%) are winners with at least 1,000 views. The test split shows a very similar distribution (42.3% non-winners vs. 57.7% winners), so there is no severe class imbalance.

As summarized in Table I, the Shorts dataset is moderately but not severely imbalanced. We later apply inverse-frequency class weighting; see Section IV-E.

For each Short, we collected:

- video-level metadata: title, description, tags;
- the thumbnail image URL;
- published timestamp; and
- view count within a fixed observation window.

Following the Shorts report, we defined a binary target: *Winner* if a Short reached at least 1,000 views within the window, and *Non-winner* otherwise.¹

1) Preprocessing for the Shorts dataset:

- Dropped Shorts with missing thumbnails or corrupted text fields;
- Normalized timestamps to Eastern Time and derived hour-of-day, day-of-week, and weekend flags;
- Ensured no target leakage by freezing the observation window before the view-count cutoff.

E. Pipeline and reproducibility

When working with YouTube’s trending dataset, we encounter a classic case of survivorship bias, as the dataset is dominated almost exclusively by viral content. Consequently, it does not capture the broader population of uploaded videos, particularly those that fail to gain traction. To systematically study the characteristics associated with video success, it is therefore essential to include both popular and unpopular content in the analysis.

To address this limitation, we adopted a stratified sampling approach in which videos were scraped across multiple predefined tiers based on view counts. This methodology was intended to ensure representation across the full spectrum of performance levels, from low-visibility uploads to highly successful videos, within the space niche. By doing so, we aimed to construct a dataset that more accurately reflects the underlying distribution of content performance.

¹Threshold approximately matches the lower bound of sustainable growth for our niche channel.

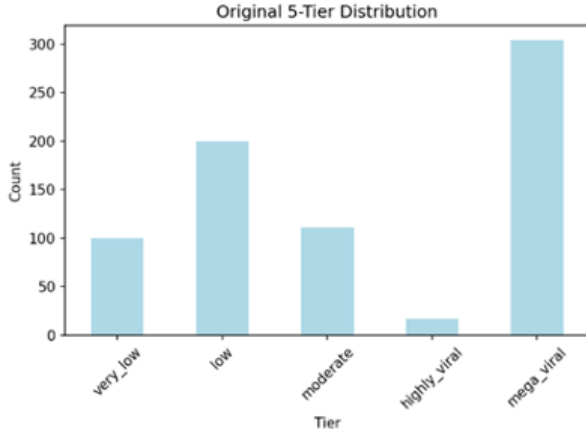


Fig. 1. Tiered Distribution

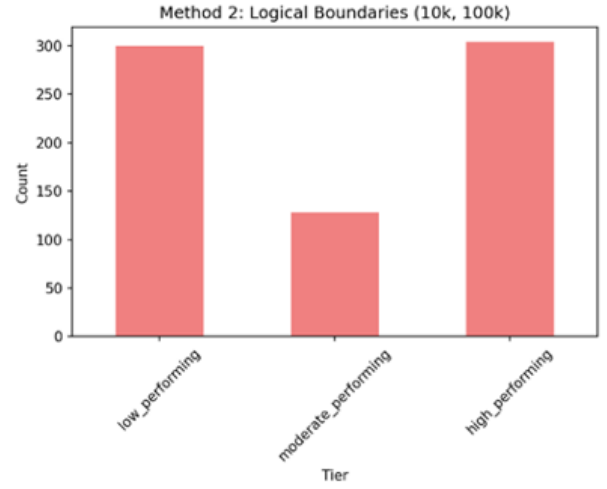


Fig. 3. Tiered Distribution

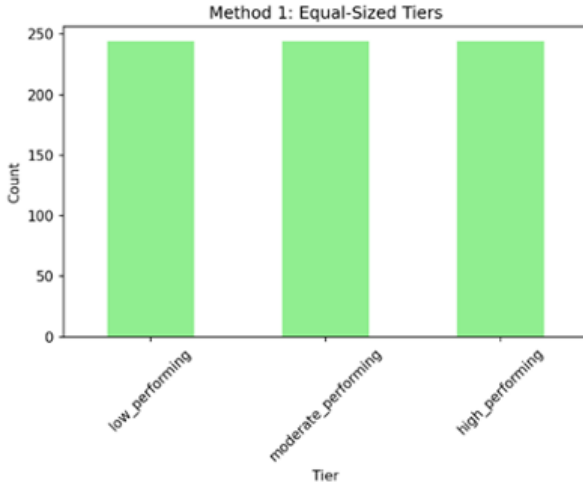


Fig. 2. Tiered Distribution

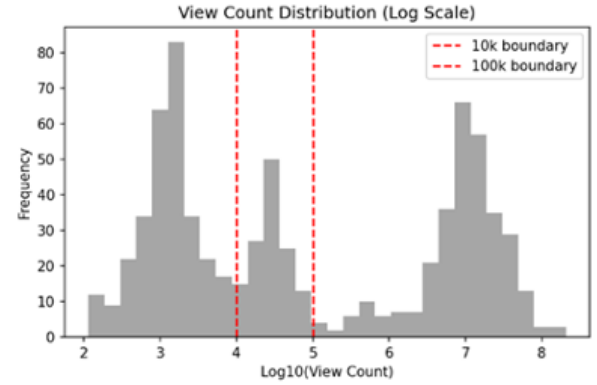


Fig. 4. Tiered Distribution

Our analysis revealed a pronounced imbalance in the availability of data across performance tiers. While video counts were high at both extremes, including very low and very high view counts, the intermediate range, specifically videos with 10,000 to 100,000 views, was severely underrepresented. This resulted in an extremely imbalanced dataset, limiting the reliability of any statistical inferences related to moderately successful videos.

We considered mitigating this issue by redefining tiers to contain equal numbers of samples per group. However, such a restructuring would artificially distort the underlying data distribution and obscure meaningful structural patterns inherent to the space niche. In fact, the data exhibits a clear bimodal distribution, reflecting natural clustering rather than deficiencies in data collection.

A further challenge was the difficulty of retrieving

low- and mid-tier videos through standard search mechanisms. These videos require the application of increasingly specific filters, significantly increasing API quota consumption. For this reason, we shifted our focus from stratified scraping to standard YouTube API based data collection, as implementing the stratified approach under the free quota limit of 1,000 units per day would require several days to retrieve a sufficiently large sample.

Let's look at YouTube Data API v3 findings. For our main analysis in this report, YouTube Data API with key word space and science, has 10,387 records ranging from Jan 2022 to Oct 2025. The YouTube API dataset has the following features, see Table II.

F. Cleaning and feature engineering

Missing values, outliers, normalization, encoding of categories, creation of targets (e.g., high/low perfor-

TABLE II
FEATURES OF SPACE TITLED VIDEO DATASET

Feature Name	Data Type
video_id	object
title	object
channel	object
published_at	object
view_count	int
like_count	int
comment_count	int
duration	object
duration_seconds	int
is_short	bool
video_type	object
tags	object
thumbnail_url	object
description	object
url	object

mance), time features, text features (title length, sentiment optional), tag vectors (optional).

We organized the workflow into three steps:

1) *Dataset Preparation and Cleaning*: We began by combining multiple monthly trending datasets into a single dataset. The rationale was to:

- ensure a complete and consistent dataset covering the full time period;
- identify missing values, duplicates, and inconsistent formats; and
- drop unreliable fields (e.g., `creatorOnRise`, which was missing in most rows).

We converted invalid durations (≤ 0 or -1) into missing values. This ensured we worked with clean, reliable data before drawing conclusions.

2) *Temporal Analysis (When to Post)*: Next, we analyzed posting times to understand how timing affects video performance. The rationale was to:

- convert video timestamps into local time (ET) for consistency;
- create hour/day features to compare publishing windows; and
- use medians and quantiles of views (instead of means) to reduce distortion from viral outliers.

We also grouped results by meaningful segments (e.g., all videos, long-form vs. Shorts, verified vs. unverified) to test interactions between timing, format, and channel authority.

3) *Title and Description Analysis (What to Say)*: Finally, we analyzed textual features of titles and descriptions:

- a) Measure lengths (characters/words) to estimate readability effects.
- b) Track punctuation and symbols (e.g., “?”, “!”) to evaluate style choices.
- c) Identify common keywords in titles/descriptions to characterize language trends.

TABLE III
KEY DATASETS USED

Dataset	Granularity	Notes
YT Analytics export	Video/day	Views, watch time, CTR, retention
YT Data API	Video	Title, publish time, tags
Google Trends	Keyword/day	External demand signal

- d) Approximate sentiment/tone (positive, neutral, negative) to explore associations with outcomes.

Each sub-step isolates one type of textual signal so it can later be combined into a predictive model. We used Python, Pandas, and Jupyter Notebook to generate the results.

III. ANALYSIS METHODS

A. Exploratory data analysis (EDA)

Describe descriptive statistics, cohorts, distributions, seasonality, correlation checks, and early hypotheses.

We organized the workflow into five steps:

1) *Dataset Overview and Interpretation Results*:

The dataset covers 13,018 trending videos (Oct. 2023–Sep. 2025) across approximately 4,000 channels.

- Verified channels dominate (71%); non-verified channels must optimize other levers.
- Shorts represent only 4.5% of trending videos; treat Shorts as a separate ecosystem.
- Views are heavily right-skewed (median $\approx 543K$ vs. mean $\approx 1.9M$); report medians and/or use a log scale.
- Long-form “sweet spot”: 8–15 minutes; Shorts ($< 60s$) rely heavily on completion rate.
- The dataset is robust with minimal missingness (except the dropped `creatorOnRise` field).

2) *Temporal Analysis (Posting Times)*: Best posting windows (ET) were (see Fig 5):

- Overall: Sat./Sun. afternoons (15–16h); Thu. 15h is reliable; Fri./Sat. 6 AM appears strong but in fewer cases.
- Long-form: weekend afternoons and Thu. 15h are robust.
- Shorts: early mornings (6 AM, especially Friday) perform best; afternoons appear oversaturated.
- Verified channels perform well in mornings and evenings; unverified channels perform better in afternoon slots (Thu./Fri./Sat./Sun. 15–16h).

Key takeaways:

- Weekends matter; afternoons are safer for unverified channels.
- Shorts are volatile; prioritize testing early-morning slots.
- There is a large authority gap: verified channels pull millions; unverified channels benefit from precise timing choices.

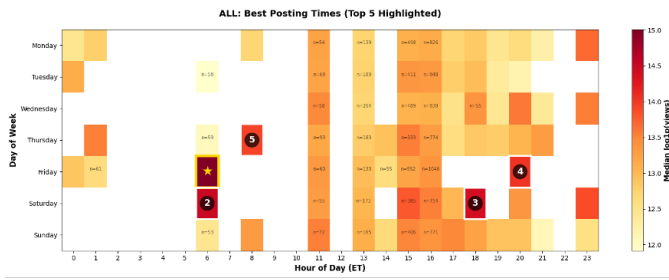


Fig. 5. Example graph of Best Posting Times for All videos regardless of format or authority (Kaggle Dataset)

3) *Title and Description Lengths*: Trending titles are typically 6–11 words (≈ 50 characters), balancing scanability with clarity. Descriptions are approximately 170 characters (about 2–3 sentences) and are often truncated around 200 characters, commonly used for links and keywords. Strategy: keep titles concise and use descriptions for keywords and links.

4) Punctuation and Symbol Use (review Fig 6):

- Approximately 5% of titles use “?”; 15% use “!”; 35% use numbers; and 6% use emojis.
- Numbers are effective for lists and challenges.
- “!” adds urgency, “?” adds curiosity, and emojis fit casual/Shorts-oriented content.

Strategy: use numbers frequently; reserve “!” for high-energy topics; use “?” sparingly; use emojis mainly for Shorts niches.

5) *Keywords in Titles and Descriptions* (see Fig 7): Common title keywords include *official*, *new*, *best*, *vs.*, *trailer*, *highlights*, *season*, and *episode*, suggesting that recency, exclusivity, and comparisons perform well. Descriptions are primarily functional (links and calls-to-action such as *subscribe*, *follow*, *merch*, *free*, *join*). Strategy: treat titles as hooks and descriptions as functional hubs (links + CTAs + SEO reinforcement).

B. Visualization

Throughout the document we have included most relevant visuals for each section with additional details added to Appendix ??

C. Modeling approach

Define your predictive tasks:

- Regression (e.g., predict views / watch hours in 24h or 7d)
- Classification (e.g., top-quartile video vs not)

Summarize baselines and advanced models (linear/logistic regression; tree-based models; optional time-series baseline). List metrics (RMSE/MAE/ R^2 , AUC/F1, calibration if applicable).

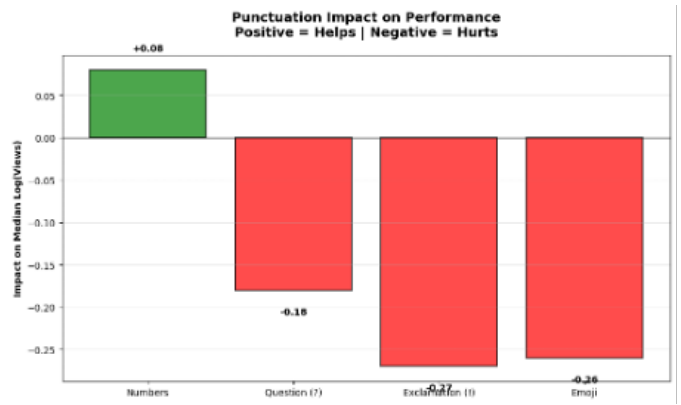


Fig. 6. Punctuation Impact (Kaggle Dataset)

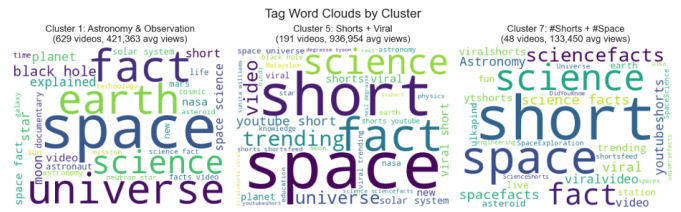


Fig. 7. Word Cloud (Space Titled Video Dataset)

D. Main Analysis - Stratified train-test split

We construct an 80/20 train-test split using equal-frequency binning of the log growth rate into ten strata. This preserves the full range of growth behavior in both sets. To verify that the split does not distort the distribution, we run Kolmogorov–Smirnov tests comparing train and test samples. The target variable yields a p -value of 0.6644, and six key features (days_since_published, duration_minutes, like_ratio, engagement_ratio, title_length, description_length) all have $p > 0.05$, confirming distribution similarity across splits.

E. Main Analysis - Model comparison

We evaluate seven regression algorithms: RandomForest, GradientBoosting, XGBoost, LightGBM, ExtraTrees,

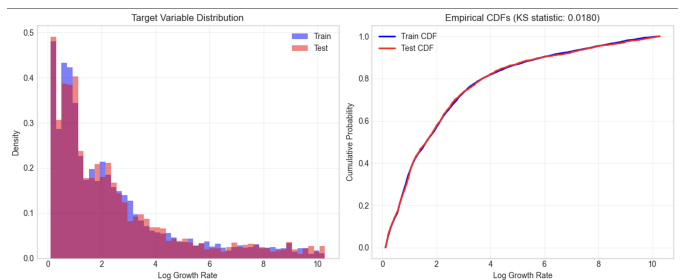


Fig. 8. Stratified train-test split

TABLE IV
BASE MODEL RESULTS ON THE YOUTUBE GROWTH DATASET.

Model	Train R^2	Test $R^2 \uparrow$	CV mean	RMSE	Time (s)
LightGBM	0.8764	0.7320	0.7062	1.2031	2.55
RandomForest	0.9576	0.7103	0.6833	1.2509	23.40
XGBoost	0.9605	0.6952	0.6520	1.2829	2.16
GradBoosting	0.7182	0.6793	0.6561	1.3159	12.56
ExtraTrees	1.0000	0.6362	0.6183	1.4016	8.54
NeuralNet	0.4275	0.3371	0.3048	1.8921	2.52
AdaBoost	0.3175	0.3182	0.2141	1.9189	2.99

TABLE V
TUNED MODEL RESULTS ON THE YOUTUBE GROWTH DATASET.

Model	Train R^2	Test $R^2 \uparrow$	CV mean	RMSE	Time (s)
XGBoost	0.9320	0.7291	0.6974	1.2096	22.90
LightGBM	0.8745	0.7278	0.7091	1.2124	24.31
GradBoosting	0.9968	0.7269	0.6979	1.2144	172.18
RandomForest	0.9373	0.7167	0.6945	1.2369	61.02
ExtraTrees	0.9999	0.6408	0.6169	1.3929	77.82
AdaBoost	0.5027	0.4774	0.4919	1.6799	50.65
NeuralNet	0.5079	0.3999	0.3688	1.8002	30.02

AdaBoost, and a feed-forward neural network. For each model we first train a baseline configuration, then apply hyperparameter optimization using Randomized-SearchCV with three-fold cross-validation over 20 sampled configurations per algorithm.

Tree-based ensemble methods consistently outperform the neural network and AdaBoost baselines. LightGBM achieves the strongest baseline performance with test $R^2 = 0.7320$, followed by RandomForest and XGBoost. For more comparison details, review TABLE IV.

F. Multimodal deep model for YouTube Shorts

For the Shorts dataset, we implemented a multimodal binary classifier (EnhancedWinnerClassifier) to predict whether a Short will be a “winner” ($\geq 1,000$ views). The model ingests three modalities:

- **Text semantics:** SentenceTransformer embeddings (384-d) for title, description, and tags;
- **Thumbnail semantics:** a 512-d CLIP vision embedding;
- **Temporal features:** six engineered time features (hour-of-day, day-of-week, weekend flag with sinusoidal encodings).

These feature vectors are processed in separate MLP “towers” and then fused through a residual block before a sigmoid output layer. Training uses AdamW, balanced BCEWithLogits loss, and early stopping on validation accuracy.

G. Classical Shorts analysis: Random Forest and clustering

Before training the multimodal EnhancedWinnerClassifier, we first analyzed a Shorts-only dataset of 9,011 space/science Shorts (2022–2024) using simpler features and classical methods [1]. The feature set included title length, binary indicators for emojis, hashtags, and questions in the title, number of tags, and duration in

seconds. A Random Forest regressor ($n_{\text{estimators}} = 200$, max depth = 5) achieved a test R^2 of 0.155, indicating that these shallow metadata features capture only a small fraction of the variance in log-transformed views. We also applied k -means clustering ($k = 4$) on the same feature space to identify Shorts archetypes and relate them to average view counts.

H. Experimentation / A/B testing design

Result from modeling provides actionable insights by enabling A/B testing (e.g., comparing title variants by percentile score). We experimented with holiday theme, and title changes such as question vs statement, number, punctuation, and emoji. For posting time, the best way for the algorithm to not flag the two A/B version videos is to make slight changes to the video while keeping the content similar. In addition, posting time need to be staggered on different weekdays at least one day apart at the same time to prevent competition between videos on the same channel. Thumbnails is not under the direct control of the content creator as this is automated by YouTube, however we do keep the parameters the same as much as we can while preventing the videos being labeled as duplicates.

1) *A/B Testing Design:* To understand how specific creative choices affect the performance of YouTube Shorts, we conducted a series of paired A/B experiments. Each experiment consisted of two Shorts (Variant A and Variant B) built around the same underlying topic and media assets (e.g., NASA images of a specific nebula, supernova, or Mars rover). The goal was to isolate the impact of a single creative lever while holding other production factors as constant as possible.

For each A/B pair, the following elements were controlled:

- **Content theme and assets:** Both variants used the same astronomical object or mission and the same pool of NASA imagery.
- **Format:** Vertical 9:16 Shorts, similar total duration (< 25 seconds), and comparable background music.
- **Language and call-to-action:** English narration or on-screen text with similar informational intent and tone.
- **Upload conditions:** Both variants were uploaded on the same channel within a narrow time window (same day and similar time of day) to reduce confounding from temporal or audience effects.

Across the three most recent A/B experiments, we systematically varied one primary factor per pair:

- **Hook and on-screen text:** In some experiments, Variant B used a more curiosity-driven or emotionally charged opening line and first 3 seconds of text, while visuals and length remained nearly identical to Variant A.
- **Initial lead image and image order:** In other cases, the sequence of images was altered so that one

variant began with a more visually striking frame, while the script and metadata were kept constant.

- **Title and description framing:** For metadata-focused tests, both variants shared identical video content, but differed in title style (descriptive/state-ment vs. curiosity-driven/question) and corresponding description phrasing, with hashtags held largely consistent.

A seasonal experiment was also conducted around Halloween. Here, the treatment was a *seasonal framing*: Variant B adopted a “spooky” or eerie narrative style and Halloween-themed wording, while retaining the same core space imagery and Short format as non-seasonal content. This test extended the same A/B framework to examine whether holiday framing affects engagement for science-focused Shorts.

Evaluation of each A/B pair relied on early performance and engagement metrics:

- **Early views and impressions** in fixed windows (approximately 1 hour, 24 hours, and up to 7 days after upload).
- **Engagement metrics**, including average view duration, percentage viewed, and retention patterns across the Short.
- **Secondary outcomes**, such as likes, comments, and subscribers gained per video, when available.

For each experiment, the variant that consistently outperformed its counterpart on early views and engagement metrics was designated the winner. Design patterns from winning variants (e.g., hook style, hero image order, or title framing) were then propagated into subsequent content production. In this way, the A/B testing procedure functioned as an iterative optimization loop, gradually refining Shorts design based on empirical evidence.

IV. RESULTS AND INTERPRETATION

A. Model performance and tuning effects

Table IV summarizes baseline performance. LightGBM attains the highest test R^2 (0.7320) with relatively modest training time, followed by RandomForest and XGBoost. Neural networks and AdaBoost underperformed substantially.

After hyperparameter tuning (Table V), GradientBoosting and XGBoost gain the most, with test R^2 improvements of approximately +7.0% and +4.9% respectively relative to their baselines. In contrast, LightGBM exhibits a slight performance decrease of 0.6% (from 0.7320 to 0.7278), illustrating a tuning paradox: the default LightGBM configuration is already near-optimal, and aggressive tuning may overfit cross-validation folds. We therefore select the *untuned* LightGBM model as our final regressor, balancing predictive power and generalization with a test R^2 of 0.7320 and an overfitting gap of 0.1444.

B. Main - Feature importance and drivers of growth

In the final LightGBM model, engagement metrics dominate feature importance. Comment ratio emerges as the single most influential feature (importance score 342), followed by days since published (254), confirming that both viewer interaction and video age are critical drivers of growth. TF-IDF components collectively carry the largest total importance (1308), indicating that the semantic content of titles and descriptions also plays a substantial role in performance.

C. Main - Text formatting and description strategy

Text analysis yields counterintuitive findings for title formatting. Traditional attention-grabbing elements such as colons, hyphens, pipes, and brackets show consistent negative correlations with log growth rate, suggesting that cleaner, more straightforward titles perform better for educational space/science content. Description analysis reveals strong positive correlations for professional elements: including email addresses, social media mentions, and URLs is associated with roughly 55%, 39%, and 25% higher log growth rates respectively. Long-form descriptions with multiple calls to action perform best, positioning the video as part of a broader professional content hub.

D. Main - Tagging and content positioning

Tag-category analysis shows that educational tags are positively associated with growth (approximately +17.2%), whereas direct space/astronomy and science/research tags are slightly negative (around −2.8% to −5.1%), despite the dataset focusing on space/science. This pattern suggests saturation in narrow science tags and highlights the benefit of framing scientific content as educational. The most effective strategy uses around five distinct tag categories, balancing topical diversity with relevance.

E. Shorts Winner-Prediction Multimodal Results and Insights

On a held-out validation set, the Shorts model achieves approximately 65.6% accuracy and 66.3% recall on the winner class, with a substantial gap between training and validation performance. This is consistent with prior work showing that content-performance prediction is inherently noisy and subject to shifting viewer preferences.

Beyond accuracy, we focus on interpretability. A three-phase framework—ablation, SHAP and permutation importance, and counterfactual flip tests—consistently reveals that:

- 1) **Title semantics are the dominant driver:** removing title embeddings causes the largest accuracy and AUC drops, and SHAP ranks titles as the top global contributor.

- 2) **Thumbnails have modest global impact but high local sensitivity:** ablating thumbnails barely hurts global metrics, yet small changes in thumbnail embeddings trigger a high rate of prediction flips, indicating thumbnails behave as local tie-breakers.
- 3) **Descriptions play a moderate supporting role,** reinforcing the title and providing additional context.
- 4) **Tags and timing features contribute minimally to predictive power within this niche,** suggesting that meta tweaks and posting time matter less than core content framing.

F. Shorts-level Patterns from Classical Analysis

The Random Forest model on the Shorts-only feature set, while modest in explanatory power (test $R^2 = 0.155$), reveals clear directional trends [1]. Title hashtags emerge as the most important feature (permutation importance 0.318), followed by title length (0.263) and duration in seconds (0.205). Correlation analysis shows that title hashtags are negatively associated with log-transformed views ($r = -0.25$), and longer titles are also detrimental ($r = -0.22$), whereas duration and other indicators play smaller roles. Emojis have a weak positive correlation ($r \approx 0.13$).

K-means clustering ($k = 4$) identifies a high-performance cluster (Cluster 3) with average views around 2.1 million. Shorts in this cluster share several traits: shorter titles (mean ≈ 54 characters), *no* hashtags in the title, a higher number of tags (mean ≈ 10), and durations in the 35–45 second range. In contrast, lower-performing clusters are characterized by longer, hashtag-laden titles and fewer tags. These patterns suggest a practical Shorts recipe for the space/science niche: clean titles without inline hashtags, 8–12 backend tags for SEO, and durations near the mid-range of the Shorts window.

V. CHANNEL PERFORMANCE OUTCOMES

Tie analysis to outcomes: subscriber growth, impressions, CTR, views, watch time, retention. Show “before vs after” and trend snapshots.

Our subscribe count grew organically without addition input other than short feed promotion. The number of views increased over time to over 32k views, with 459+ likes, and 11.5k engaged views. Engaged views is where the viewer watched past the initial seconds of the video. Higher likes and viewership with Question based titles.

VI. STRATEGIES IMPLEMENTED AND EFFECTIVENESS

From our main analysis, the combined model and text analysis suggest a multi-faceted optimization strategy:

- **Maximize engagement:** prioritize content that naturally generates comments, as comment ratio is the most predictive feature.

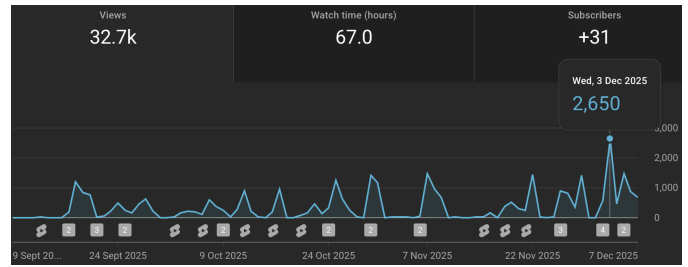


Fig. 9. Example: Subscriber growth and CTR trend over time.

- **Simplify titles:** avoid heavy formatting and use clear, direct language that matches educational intent.
- **Professional descriptions:** include contact information, social links, and multiple calls to action to signal credibility and provide engagement pathways.
- **Educational positioning:** frame space/science videos with educational tags rather than relying solely on narrow science labels.
- **Diverse tagging:** span roughly five tag categories to maximize discoverability while maintaining relevance.

Properly optimized videos following these guidelines can achieve substantial gains; our estimates suggest the potential for 50–100% higher view rates compared to sub-optimally configured videos, with the single most impactful change being the inclusion of professional contact information in descriptions.

VII. REFLECTION: LESSONS LEARNED AND CHALLENGES

A. What worked and why

Targeted training with multimodal binary classifier reveal useful information regarding Shorts. Also, having at least one person on the team with proficiency in video editing tools is invaluable in saving time for the entire team to do data analysis as videos do take a long time to learn and make. Since time is of the essence, you will see we focus solely on short videos for generation and analysis, and excluded long form videos.

B. What did not work and why

We found the Kaggle Trending dataset to be of limited use after initial analysis because of survivorship bias remove information needing for early channel analysis. Also we wanted to focus on Short form video for practical reasons.

C. Limitations and threats to validity

The sample of A/B testing was limited to 5 sets and the details of the YouTube algorithm is not publicly disclosed and we can only infer from behavior from our testing.

VIII. RECOMMENDATIONS AND FUTURE WORK

Building on the quantitative findings and Shorts-level experiments, we propose the following actionable recommendations for educational space/science creators. Each recommendation specifies an action, metric, target, and time window to enable measurable progress.

A. Actionable Recommendations

R1: Maximize comment-driven engagement.

- *Action:* Design each Short with at least one explicit discussion prompt (e.g., a question in the title or final caption) aimed at eliciting comments rather than passive views.
- *Metric:* Comment ratio (comments per 1,000 views).
- *Target:* Increase median comment ratio by 30% relative to the channel’s baseline within three months.
- *Rationale:* The LightGBM model identifies comment ratio as the single most influential feature for growth, indicating that interactive content strongly drives performance.

R2: Simplify titles and remove heavy formatting.

- *Action:* Standardize titles to short, clean sentences without colons, pipes, brackets, or inline hashtags, prioritizing clarity over formatting tricks.
- *Metric:* Proportion of new uploads with “clean” titles (no colons, hyphens, pipes, or brackets; no title hashtags).
- *Target:* Achieve at least 90% compliance on new uploads and track relative change in median log-view growth over the following 90 days.
- *Rationale:* Text analysis shows that traditional attention-grabbing punctuation consistently correlates with lower growth, while straightforward titles perform better in the educational space/science niche.

R3: Professionalize descriptions as content hubs.

- *Action:* Adopt long-form descriptions that include email contact, at least one social media handle, one external URL, and multiple calls-to-action (CTAs) per video.
- *Metric:* Count of professional elements per description (email, social link, URL, and number of CTAs).
- *Target:* For all new uploads, include at least one email, one social link, one URL, and approximately eight CTAs, and evaluate the impact on log-growth over a 3–6 month horizon.
- *Rationale:* Professional elements in descriptions (email, social media, URLs) are associated with substantial gains in log growth, and long descriptions with multiple CTAs perform best.

R4: Reframe space/science content as educational.

- *Action:* Position videos explicitly as educational content in both tags and description language, rather than relying solely on narrow space or research labels.

- *Metric:* Fraction of videos whose tag sets include an education-related category alongside space/science tags.
- *Target:* Ensure that at least 80% of uploads use educational tags and monitor changes in median log-growth over the next quarter.
- *Rationale:* Tag-category analysis shows that educational tags have the strongest positive association with growth, while pure space/astronomy and science/research tags are slightly negative, suggesting saturation in narrow science niches.

R5: Adopt a structured Shorts recipe and A/B testing loop.

- *Action:* For Shorts, follow a standardized recipe (clean titles, 25 second durations, 8–12 backend tags) and continuously run A/B tests on hooks, lead images, and title phrasing.
- *Metric:* Share of Shorts that conform to the recipe and the number of A/B experiments completed per month.
- *Target:* Bring at least 75% of Shorts into the recipe and run 2–3 A/B tests per month, selecting winners based on early views, retention, and engagement.
- *Rationale:* Cluster analysis of Shorts identifies a high-performing group characterized by mid-range durations, clean titles without hashtags, and a higher number of tags. Iterative A/B testing allows these patterns to be refined and validated in practice.

R6: Integrate insights into future personalized retrieval systems.

- *Action:* Extend the current prototype personalized retriever into a full Retrieval-Augmented Generation (RAG) pipeline that incorporates title semantics, educational framing, and engagement-driven ranking signals.
- *Metric:* Offline ranking quality (e.g., Precision@5, Recall@5) and user-level engagement for recommended videos in pilot tests.
- *Target:* Demonstrate measurable improvements in ranking metrics over a semantic-only baseline, and document qualitative gains in recommendation relevance.
- *Rationale:* The prototype shows that combining semantic similarity, popularity, and user-profile embeddings is feasible; integrating empirically validated text and tag strategies should further improve personalized discovery in future systems.

B. Personalized retrieval and RAG-style extensions

Beyond modeling what drives video growth, our teammate explored how these insights could feed into a personalized retrieval module for future Retrieval-Augmented Generation (RAG) systems [2]. In this prototype, video titles, descriptions, and tags from the same

space/science collection are encoded with a Sentence-BERT model and indexed for semantic search. A personalized re-ranking layer then combines three signals: semantic similarity between query and video, a log-scaled view-count popularity term, and a personalization score based on the cosine similarity between a user’s profile embedding and candidate videos.

Experiments compare three ranking strategies using Precision@5 and Recall@5 on a small set of hand-labeled queries: a pure semantic baseline, a view-count re-ranked variant, and a hybrid personalized mode. On this evaluation set, the semantic baseline performs best on average (Precision@5 $\approx$ 0.55, Recall@5 = 1.0), while both popularity and personalization variants slightly degrade these metrics, illustrating how easy it is to harm a strong baseline when adding additional signals. At the same time, qualitative case studies show that the personalized variant does behave like a recommender: two simulated users issuing the same query (e.g., “space news”) receive clearly different, preference-aligned rankings (black-hole-focused results versus exoplanet-focused results).

This work is therefore best viewed as a proof of concept for a future RAG-style extension rather than a core component of our current pipeline. Promising directions include: (i) replacing handcrafted user profiles with sequence-aware models that track viewing behavior over time; (ii) learning query- and user-specific weights for personalization instead of using a single global parameter; and (iii) integrating the personalized retriever into a full RAG system where the top- k retrieved videos provide grounding for an LLM that generates user-tailored, space/science recommendations.

IX. CONCLUSION

Using a dataset of 10,387 YouTube videos (Jan. 2022–Oct. 2025) for main analysis and 9,247 space/science Shorts (Jan. 2022–Oct. 2025) collected via the YouTube Data API [3], we built an end-to-end analytics workflow for discovery, growth forecasting, and channel optimization for *SpaceMSDA*. Among the evaluated models, LightGBM delivered the strongest generalization for predicting log-normalized view growth (test $R^2 \approx$ 0.73). Across model families, interpretability consistently indicated that engagement metrics (especially comment ratio), time effects (days since published), and text semantics in the title were the primary drivers of growth. These results are directionally consistent with public descriptions of YouTube’s recommendation and discovery systems, which are personalized and learn from large numbers of behavioral and content signals [4], [5], [10].

Operationally, we translated the analysis into repeatable creator actions: explicit comment prompts to induce discussion, simplified and standardized title templates for clarity, and disciplined online A/B tests of video choices following established experimentation

practice [11]. We also intentionally de-emphasized tags relative to higher-impact metadata elements such as titles and descriptions, consistent with YouTube’s own guidance that tags play a limited role for most videos compared with core metadata [7]. Key limitations include non-stationary viewer preferences, sampling bias from keyword-based collection, and shifting platform measurement—notably YouTube’s March 31, 2025 update to how Shorts “views” are counted (with “engaged views” retained separately) [8]. Future work will incorporate retention- and impression-based metrics (e.g., CTR and audience retention) [6], automate experiment tracking, and test causal uplift of interventions via ongoing controlled experiments.

ACKNOWLEDGMENT

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APPENDIX A

SUPPLEMENTARY FIGURES AND TABLES

A. YouTube Shorts model figures

Review Fig. 10 to Fig. 15 for details of the Short Model.

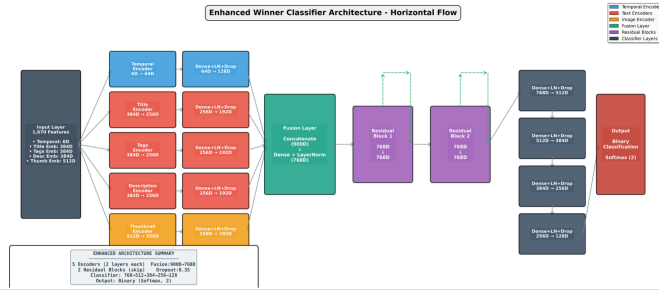


Fig. 10. Architecture of the multi-modal Shorts winner-prediction model (EnhancedWinnerClassifier).

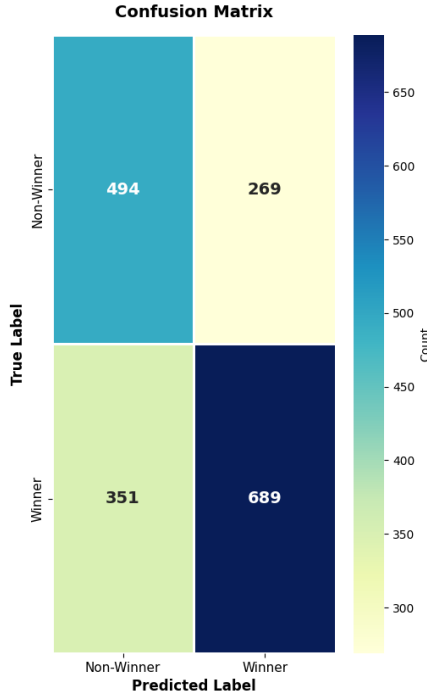


Fig. 11. Confusion matrix the Shorts EnhancedWinnerClassifier Model.

APPENDIX B MODEL AND EXPERIMENT DETAILS

A. Shorts Model Architecture

We implement an EnhancedWinnerClassifier composed of separate towers for title, description, tags, thumbnail, and temporal features. Each modality-specific tower is a two-layer MLP with a Dense-ReLU-Dropout stack, producing a compact representation for that modality.

The tower outputs are concatenated and passed through a fusion block (Dense-ReLU-LayerNorm-Dropout), followed by a residual block consisting of another Dense-ReLU layer with an add-and-normalize skip connection. A final sigmoid classification layer outputs the probability that a Short is a winner (view count $\geq 1,000$).

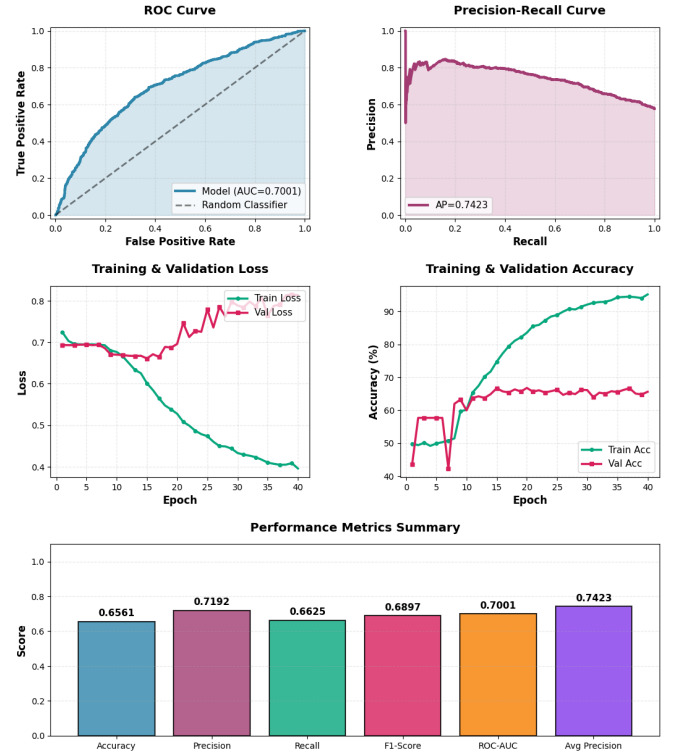


Fig. 12. Training and validation accuracy for the Shorts EnhancedWinnerClassifier Model.

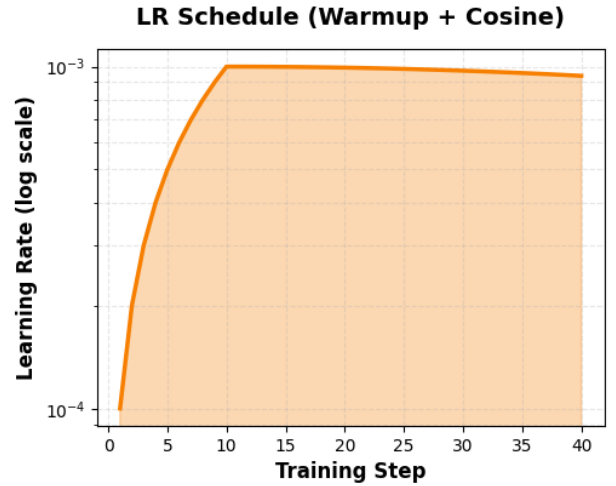


Fig. 13. Learning Rate for the Shorts EnhancedWinnerClassifier Model.

Figure 10 illustrates the overall architecture of the EnhancedWinnerClassifier.

B. Shorts Deep Learning Training Setup

We train the EnhancedWinnerClassifier on mini-batches of Shorts examples with a balanced loss:

- Optimizer: AdamW with initial learning rate 10^{-3} and cosine-annealed learning rate schedule;
- Loss: class-weighted BCEWithLogitsLoss to compensate for the mildly skewed label distribution;
- Class weights: $w_0 = 1.1906$ for non-winners and $w_1 = 0.8620$ for winners (computed from the training-set frequencies; class-balance ratio ≈ 0.72);
- Training batches: 113 mini-batches per epoch;
- Test batches: 29 mini-batches for evaluation.

The class weights follow the standard inverse-frequency scheme $w_c = \frac{N}{2N_c}$, where N is the total number of training examples and N_c is the number of examples in class c .

C. Shorts Interpretability Protocol

We structure interpretability into three phases: Phase 1 (modality ablations and gradient sensitivity), Phase 2 (SHAP and permutation importance), and Phase 3 (counterfactual flips and embedding concentration).

Phase 1 focuses on *local, model-specific importance*: methods that operate inside the trained network and quantify the immediate effect of changing or removing each modality on the prediction. These techniques rely on the model’s internal weights, gradients, and forward passes and answer the question:

“If we modify or remove this modality inside the model, how much does the prediction change?”

1) Phase 1: Local model-specific importance:

a) *Ablation study: understanding end-to-end dependence*: Ablation removes a modality entirely (e.g., setting title embeddings to zero) and re-trains or re-evaluates the model to measure the drop in predictive performance.

What it measures: ablation directly quantifies how much the model relies on a given modality to make correct predictions.

Why it matters: a large accuracy drop indicates that the modality is essential, whereas a small drop suggests that the model can function without it.

Interpretation:

- A large accuracy drop $\Rightarrow$ modality is critical.
- A small drop $\Rightarrow$ modality is supportive or redundant.

Our results:

- Title removal $\rightarrow$ 7.95% accuracy loss (critical);
- Description removal $\rightarrow$ 2.70% loss (moderate);
- Thumbnails, tags, temporal features $\rightarrow \approx 0.2\%$ loss (minor).

These results reveal a fundamental dependency of the model on semantic title information.

b) Gradient sensitivity: local change sensitivity:

Gradient-based analysis measures how much the prediction changes when each input dimension is perturbed

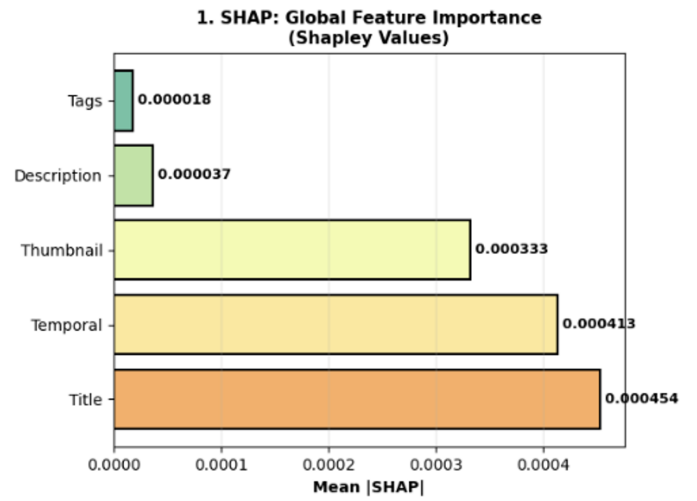


Fig. 14. Global SHAP importance for Shorts modalities.

slightly, using the magnitude of the gradient with respect to the input features.

What it measures: gradients capture local sensitivity of the output to small changes in each modality.

Why it matters: even if a modality is not globally important in terms of overall accuracy, it may still strongly affect decision boundaries or reveal fragile behavior.

Interpretation:

- Large gradients $\Rightarrow$ predictions change sharply with small input perturbations;
- Small gradients $\Rightarrow$ predictions are stable and relatively insensitive to that modality.

Our results:

- High gradient sensitivity for thumbnail and temporal features;
- Moderate gradients for titles;
- Low gradients for tags and descriptions.

This helps explain why thumbnails can trigger many prediction flips in counterfactual tests, even though ablating the thumbnail modality causes only a small global accuracy drop.

2) *Phase 2: Global model-agnostic importance*: Phase 2 focuses on *global, model-agnostic importance*: methods that treat the model as a black box and evaluate importance by manipulating the inputs rather than inspecting internal weights. These approaches answer the question:

“If we disrupt this modality in the dataset, how much does the model’s overall performance degrade?”

By operating at the population level, these methods provide a global view of how much each modality contributes to predictive performance.

a) *SHAP values: fair attribution of global contribution*: *What it measures*: SHAP values quantify how much each modality contributes to the model output relative

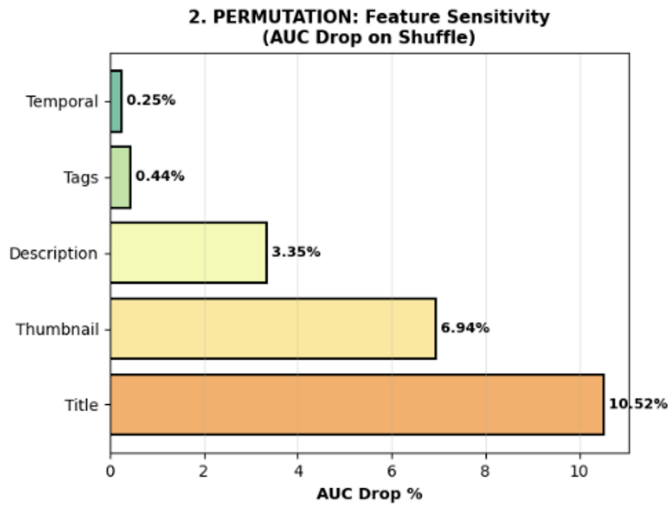


Fig. 15. Permutation importance across modalities for the Shorts model.

to a baseline, aggregating local attributions over many examples.

Why it matters: SHAP offers:

- fair, game-theoretic attributions of contribution;
- direction-agnostic global importance scores; and
- a unified explanation format for multimodal inputs.

Interpretation: the global SHAP ranking for our Shorts model is (see Fig. 14):

- 1) Title (dominant);
- 2) Temporal features;
- 3) Thumbnail;
- 4) Description;
- 5) Tags.

SHAP thus confirms the dominance of title semantics and shows that thumbnails have modest but non-zero global importance.

b) Permutation importance: robustness under destruction: What it measures: permutation importance randomly shuffles a modality across examples, breaking its relationship to the target while preserving its marginal distribution, and then measures the resulting drop in performance.

Why it matters: this is a strong model-agnostic importance measure because:

- it reveals the actual drop in predictive power when a modality is destroyed;
- it naturally handles nonlinear interactions;
- it tests sensitivity at the full-dataset level.

Interpretation (AUC drops) (see Fig.15):

- Title → 10.52% AUC drop (largest global impact);
- Thumbnail → 6.94% drop (secondary impact);
- Description → 3.35% drop;
- Tags and temporal features → nearly negligible changes.

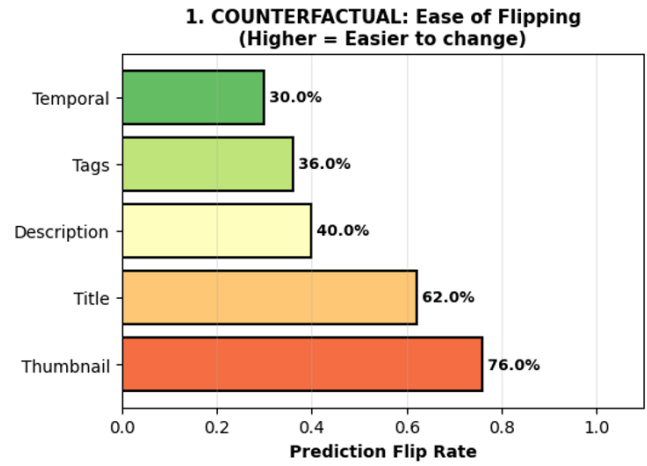


Fig. 16. Phase 3 Counterfactual

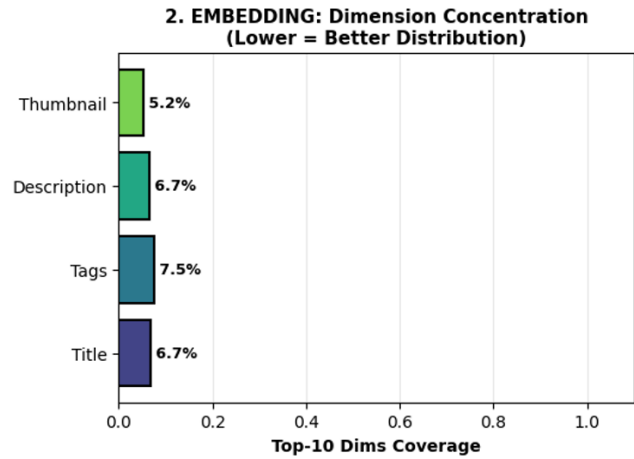


Fig. 17. Phase 3 Embeddings

These results provide strong global confirmation that titles matter most for Shorts performance, while thumbnails play a moderate but consistent supporting role.

3) Phase 3: Advanced diagnostics: Phase 3 goes beyond standard importance scoring and focuses on model stability, decision boundaries, and embedding structure. It addresses the question:

“Even if a modality is not globally important, does it destabilize decisions or distort the embedding space?”

These diagnostics capture phenomena that ablation, SHAP, and permutation importance cannot fully reveal on their own (see Fig. 16 to Fig. 21).

a) Interaction analysis (summary): Beyond single-modality diagnostics, we also inspected pairwise interaction effects (e.g., SHAP-style interaction indices). We

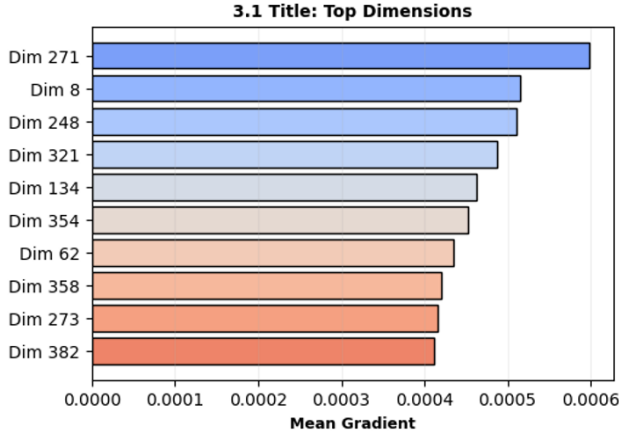


Fig. 18. Phase 3 Title Dimension

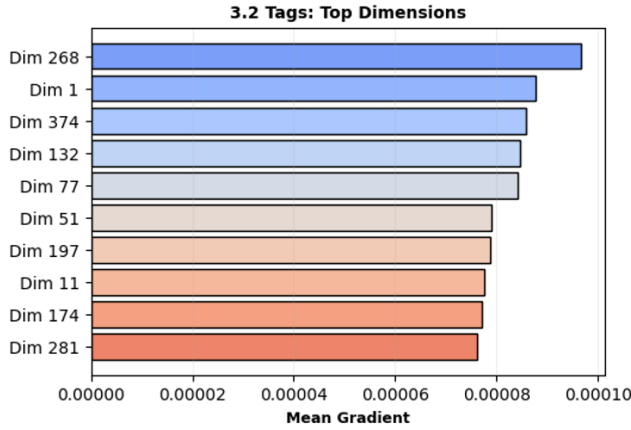


Fig. 19. Phase 3 Tags Dimension

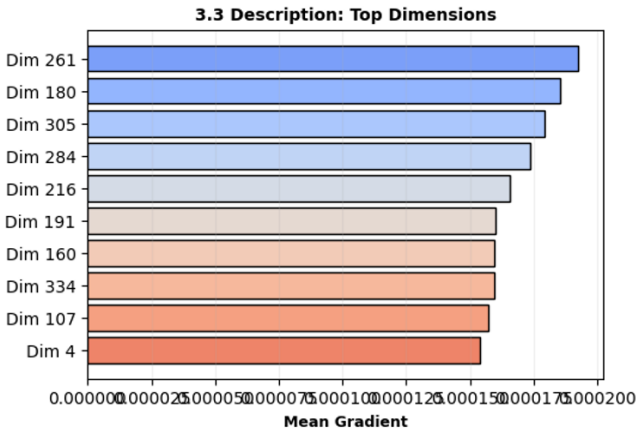


Fig. 20. Phase 3 Description Dimension

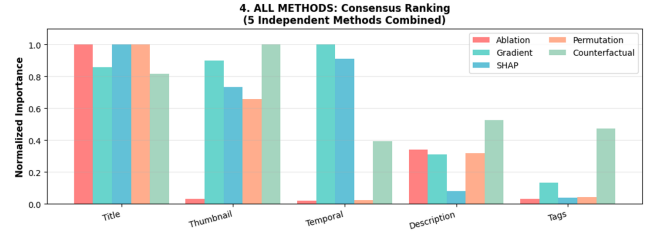


Fig. 21. Phase 3 Consensus Ranking of 5 Methods

did not observe strongly synergistic modality pairs. The main pattern is a mild redundancy between titles and thumbnails (interaction score -0.0144), indicating that thumbnails largely overlap with title information rather than introducing a wholly distinct signal. This is consistent with our Phase 3 findings: thumbnails rarely change global performance metrics but often act as local tie-breakers for borderline examples.

b) Counterfactual flip testing: decision-boundary behavior: What it measures: counterfactual flip testing perturbs each modality slightly while holding the others fixed and checks whether the model's prediction changes (flips) from winner to non-winner or vice versa.

Why it matters: this evaluates local decision-boundary stability rather than global influence on aggregate metrics.

Our results (flip rates):

- Thumbnail perturbations: 76% of cases flip;
- Title perturbations: 62% of cases flip;
- Other modalities: significantly lower flip rates.

Interpretation:

- Thumbnails do not strongly improve global accuracy (low ablation and low permutation importance),
- yet they strongly affect which side of the decision boundary a given Short falls on.

This reconciles thumbnails being *locally powerful* but *globally weak*: they rarely change overall performance but often act as tie-breakers for borderline examples.

c) Embedding concentration: structural semantics: What it measures: embedding concentration analysis examines whether a small subset of dimensions in each embedding dominates the variance or signal.

Why it matters:

- overly concentrated embeddings may cause the model to overfocus on a few dimensions, increasing brittleness;
- more diffuse embeddings indicate a more balanced semantic representation across dimensions.

Our results:

- Title, tags, and description embeddings each show roughly 6–7% coverage by their most influential dimensions;

- Thumbnail (CLIP) embeddings show 5.18% coverage.

This indicates that CLIP thumbnail embeddings are slightly more dimension-concentrated, which aligns with their high flip sensitivity in counterfactual tests.

D. Main Analysis

Using the space & science dataset (scrapped from YouTube Data API with key word space and science, has 10,387 records ranging from Jan 2022 to Oct 2025), we design a pipeline to reverse-engineer YouTube’s recommendation algorithm through predictive modeling. This technical documentation presents a comprehensive machine learning pipeline designed to analyze and predict YouTube video growth patterns. The pipeline employs advanced feature engineering techniques, multiple machine learning algorithms, and sophisticated text analysis to understand the factors driving video performance. Through systematic data processing, model training, and hyperparameter optimization, the system achieves a test R^2 score of 0.7320 (LightGBM), indicating strong predictive capability. The analysis focuses particularly on videos related to space and science content, providing actionable insights for content creators seeking to optimize their video performance.

The pipeline integrates seven distinct machine learning algorithms, ranging from tree-based ensemble methods to neural networks, enabling comprehensive model comparison and selection. A particular emphasis is placed on text feature analysis, examining how titles, descriptions, and tags influence video growth rates. The system processes video records through multiple stages, including data cleaning, feature engineering, stratified sampling with distribution validation, model training, hyperparameter tuning, and comprehensive insight generation.

1) *Data Pipeline:* The pipeline architecture consists of several interconnected modules, each serving a specific purpose in the analysis workflow (see Fig. 22). The Data Preparation module handles initial data loading and cleaning operations, ensuring data quality through systematic validation and preprocessing. The Feature Engineering module transforms raw video metadata into 92 engineered features across temporal (10 features), video-specific (4 features), textual (55 features), engagement (3 features), and TF-IDF (100 features, reduced to 20 after SVD reduction) categories. The Model Training module manages the training and evaluation of seven different machine learning algorithms (RandomForest, GradientBoosting, XGBoost, LightGBM, ExtraTrees, AdaBoost, and NeuralNetwork), implementing both baseline and hyperparameter-tuned versions.

The pipeline incorporates a sophisticated stratification mechanism that ensures representative train-test splits while maintaining distribution similarity, validated through Kolmogorov-Smirnov statistical tests. The

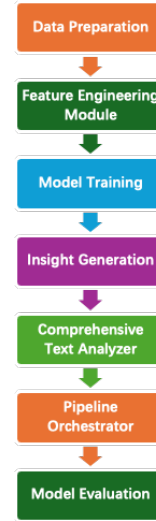


Fig. 22. Data Pipeline

Insight Generation module synthesizes model outputs into interpretable findings, while the Comprehensive Text Analyzer provides deep analysis of textual features and their impact on video performance. The entire system is orchestrated through a central pipeline controller that manages execution flow, logging, and result persistence.

2) Feature Engineering and Target Variable:

- **92 Engineered Features:** Created across 5 categories (temporal, video, text, engagement, TF-IDF).
- **Target Variable:** Log-transformed growth rate, $\log(1 + \text{views_per_day})$, aligns with YouTube’s focus on growth, reduces skewness, and stabilizes variance across the prediction range.
- **Text Analysis:** 20+ text features, including pattern detection, keyword counting, and structural metrics.
- **Dimensionality Reduction:** TF-IDF features reduced from 100 to 20 components via SVD, capturing $\sim 47\%$ of the variance.

The feature engineering module transforms raw video metadata into 92 carefully crafted features designed to capture diverse aspects of video performance potential. Temporal features extract multiple time-based components from publication timestamps, including days since publication, publication hour, day of week, month, and quarter. The module creates binary indicators for weekend publications and time-of-day categories (morning, afternoon, evening, night), enabling the model to capture temporal patterns in viewing behavior.

Video-specific features focus on content characteristics, particularly duration-based attributes. The pipeline categorizes videos into short (under 60 seconds), medium (60–600 seconds), and long (over 600 seconds) formats, reflecting YouTube’s content taxonomy. A log-transformed duration feature addresses the skewed dis-

tribution of video lengths, improving model performance on duration-related patterns.

Text feature engineering represents a sophisticated component of the pipeline, extracting over 20 features from titles and descriptions. Title analysis includes length metrics, word counts, capitalization ratios, and pattern detection for questions, exclamation marks, emojis, numbers, and special characters. The pipeline implements trending keyword detection, counting occurrences of high-impact terms identified through domain analysis. Description features encompass URL counts, hashtag usage, call-to-action patterns, and structural metrics. Tag processing includes categorization into six thematic groups (education, technology, space/astronomy, science/research, entertainment, news/documentary), enabling analysis of content positioning strategies.

The selection and transformation of the target variable represent a critical technical decision that substantially impacts model performance. The target `log_growth_rate` is a compound transformation that normalizes temporal effects of video views and emphasizes the dynamics of view changes rather than absolute view count. It is defined as:

$$\text{log_growth_rate} = \log \left(\frac{\text{view_count}}{\text{days_since_published}} + 1 \right), \quad (1)$$

where the addition of 1 prevents undefined values for videos with zero views per day. This formulation aligns with YouTube’s algorithmic preference for growth over size. Empirically, the transformation reduces the skewness of the target distribution from 18.7 to 0.3, effectively converting multiplicative relationships into additive ones that linear models can better capture. It also stabilizes variance across the prediction range, preventing the model from being dominated by a small number of extremely successful videos.

3) Model Training and Hyperparameter Tuning:

- **Stratification Method:** 10 equal-frequency bins based on log growth rate, ensuring an 80/20 train-test split.
- **Distribution Validation:** Kolmogorov–Smirnov tests confirm statistical similarity between train and test (all p -values > 0.05).
- **Best Model:** LightGBM with base parameters (test $R^2 = 0.7320$), outperforming all tuned models.
- **Tuning Paradox:** LightGBM performance decreases by 0.6% after tuning, suggesting near-optimal default parameters.
- **Significant Improvements:** GradientBoosting (+7.0%) and XGBoost (+4.9%) benefit most from tuning.
- **Model Selection Enhancement:** Pipeline compares both base and tuned models to identify the true optimum.

The model training phase evaluates seven distinct algorithms, establishing baseline performance before con-

ducting systematic hyperparameter optimization. Initial experiments reveal LightGBM as the strongest baseline performer with a test R^2 of 0.7320, followed by RandomForest (0.7103) and XGBoost (0.6952). Tree-based ensemble methods consistently outperform other approaches, with neural networks and AdaBoost showing substantially lower baseline performance.

Hyperparameter tuning employs RandomizedSearchCV with 3-fold cross-validation, exploring 20 parameter combinations per algorithm. The optimization process yields substantial improvements for gradient boosting methods: GradientBoosting achieves a 7.0% relative improvement, and XGBoost gains 4.9% through tuning. The optimal GradientBoosting configuration uses 200 estimators, a minimum samples split of 2, a minimum samples leaf of 2, a maximum depth of 20, and a maximum feature fraction of 0.5, achieving a cross-validation score of 0.6945. In contrast, LightGBM exhibits a slight performance decrease of 0.6% after tuning (from 0.7320 to 0.7278). This counterintuitive result suggests that LightGBM’s default parameters are already near-optimal for this dataset and that aggressive tuning may induce overfitting on the cross-validation folds.

The final model selection identifies the untuned LightGBM baseline as the optimal algorithm, achieving a test R^2 of 0.7320 with an overfitting gap of 0.1444, representing an effective balance between predictive power and generalization. This outcome highlights that hyperparameter tuning does not guarantee improvement and that well-designed algorithms such as LightGBM can perform excellently with default settings. The chosen configuration maintains balanced regularization without excessive tuning, achieving the best trade-off between model complexity and generalization, with an overfitting gap of only 0.1444 (compared to 0.2029 for tuned XGBoost).

4) Key Findings:

- **Model Performance & Importance:** Comment ratio is the dominant single feature (importance: 342), while TF-IDF components are collectively most important (total: 1308).
- **Model Metrics:** Optimal LightGBM achieves test $R^2 = 0.7320$ with a modest overfitting gap of 0.1444.
- **Temporal Influence:** `days_since_published` ranks second in importance (254), confirming the central role of video age.
- **Text Strategy:** Clean, straightforward titles and professionally structured descriptions with multiple calls-to-action yield the best growth.
- **Tag Strategy:** Educational tags outperform pure space/science tags; the best results come from using five diverse tag categories.
- **Impact Potential:** Combining all optimizations can yield an estimated 50–100% uplift in view rates.

a) Model performance and feature importance.:

The analysis reveals that engagement metrics dominate

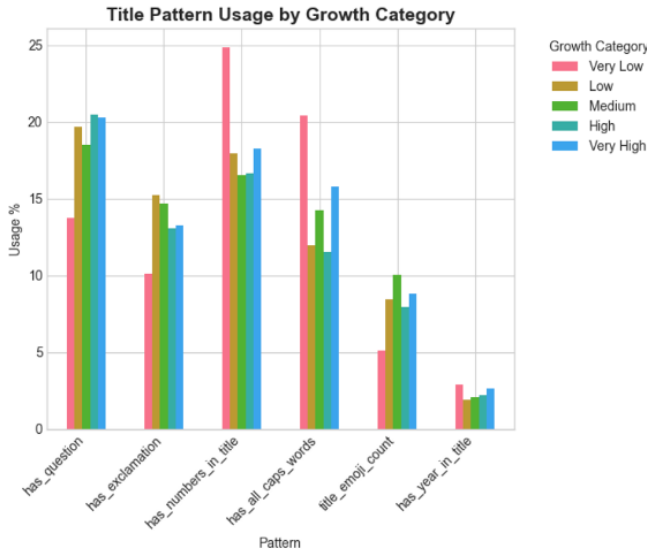


Fig. 23. Title Pattern Usage by Growth Category

feature importance in the optimal LightGBM model. Comment ratio emerges as the single most influential feature with an importance score of 342, substantially exceeding other features. This suggests that videos generating discussion and viewer interaction experience superior growth trajectories. The temporal feature `days_since_published` ranks second (254), confirming that video age remains a critical factor in growth dynamics. Notably, TF-IDF components collectively contribute the highest total importance (1308), indicating that textual content semantics play a crucial role in video performance.

b) Text analysis insights.: Comprehensive text analysis uncovers counterintuitive patterns in title formatting that challenge conventional content optimization wisdom. Traditional attention-grabbing elements demonstrate consistent negative correlations with growth: colons decrease performance by 19.6%, hyphens by 9.7%, pipes by 7.1%, and brackets by 6.9%. This pattern suggests that cleaner, more straightforward titles resonate better with audiences seeking educational or scientific content, and that heavily formatted titles may be perceived as clickbait or less credible.

Description analysis reveals substantial positive impacts from professional networking elements. Including email addresses correlates with 54.9% higher log growth rates, while social media mentions add 38.7% and URLs contribute 25.3%. These findings suggest that videos positioned as professional content hubs or as part of broader content ecosystems achieve superior growth. The optimal description length falls in the “Long” category, with comprehensive descriptions containing approximately eight call-to-action elements performing best, indicating that viewers value detailed information

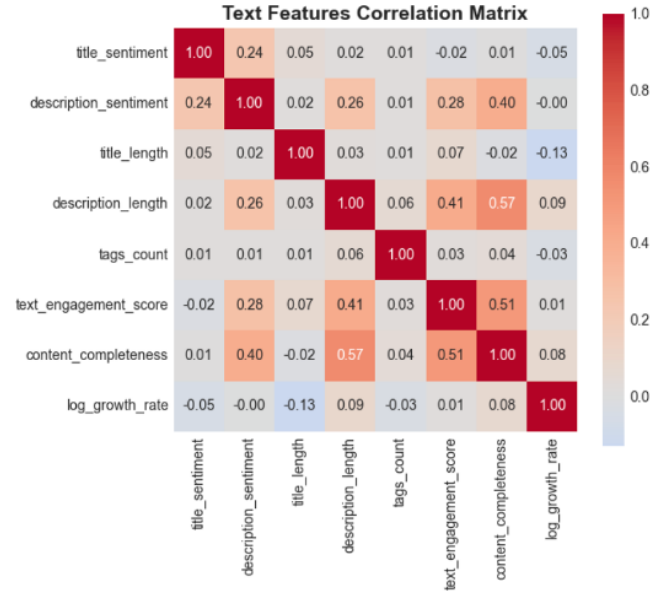


Fig. 24. Correlation Matrix

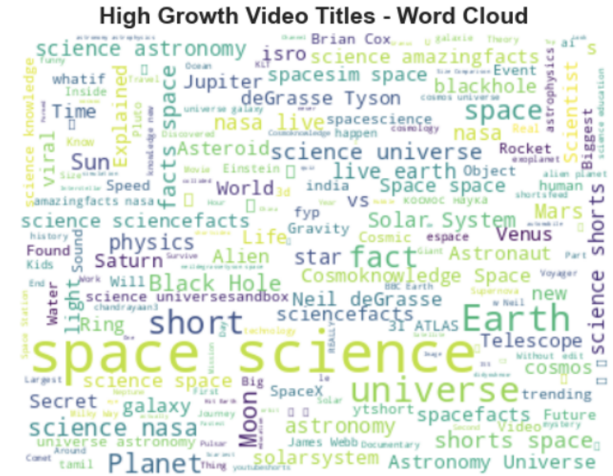


Fig. 25. Title Words for Growth

and clear engagement pathways.

c) Tag strategy findings.: Tag categorization analysis presents surprising insights for space and science content. Despite the dataset focusing on space/science videos, educational tags demonstrate the strongest positive correlation (+17.2%), while `space_astronomy` (−2.8%) and `science_research` (−5.1%) tags show negative correlations. This paradox suggests market saturation in specific science categories and indicates that framing scientific content with educational positioning improves discoverability and engagement. The optimal tag strategy employs five different category types (education, technology, `space_astronomy`, `science_research`,

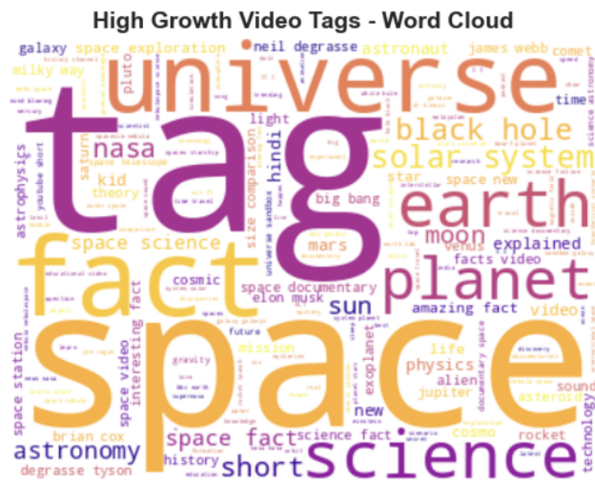


Fig. 26. Tags Words for Growth

entertainment, news_documentary), maximizing topical coverage while maintaining relevance.

d) *Actionable recommendations.*: Based on these findings, content creators should adopt a multi-faceted optimization strategy:

- 1) **Maximize engagement:** Prioritize comment-generating content, as comment ratio is the most predictive factor.
- 2) **Simplify titles:** Remove excessive formatting and use clear, direct language.
- 3) **Professional descriptions:** Include contact information, social links, and multiple calls-to-action.
- 4) **Educational positioning:** Frame science content as educational rather than purely research-focused to avoid saturated niches.
- 5) **Diverse tagging:** Use a diversified tag portfolio spanning five category types for optimal discoverability.

e) *Impact potential.*: The analysis suggests that properly optimized videos can achieve 50–100% higher view rates relative to sub-optimally configured content. The single most impactful individual change is the inclusion of professional contact information in descriptions. When combined with simplified titles, professional long-form descriptions, educational framing, and diverse tagging, these practices create compounding positive effects on growth and audience engagement.